

# Daniel T. Lacosta

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Portfolio: <https://mendo03.github.io/>

## PROFESSIONAL SUMMARY

Computer Engineering graduate with hands-on experience in systems administration, Active Directory environments, and IT infrastructure. Built and documented a home lab simulating a real enterprise domain network covering domain setup, user management, Group Policy, shared folder permissions, and common helpdesk scenarios. Experienced in hardware troubleshooting, IoT systems, and technical documentation through engineering projects and 300-hour OJT. Seeking an entry-level IT support or Systems Administration role where I can apply practical skills and continue growing in a professional environment.

## AREA OF EXPERTISE

**IT & NETWORKING:** TCP/IP Configuration | Network Troubleshooting | Device Connectivity | Ethernet Setup | Basic System Administration

**TOOLS & PLATFORM:** Windows Troubleshooting | Linux | GitHub | PuTTY | SSH | Command Line

**HARDWARE & SYSTEMS:** Raspberry Pi | Arduino | GSM Modules | PCB Prototyping | Hardware Troubleshooting

**PROGRAMMING:** Python | C++ | OpenCV | Basic Automation Scripting

**CYBERSECURITY & CLOUD FUNDAMENTALS:** AWS Cloud Foundation | CISCO Cybersecurity | Ethical Hacking Fundamentals

## WORK EXPERIENCE

### IoT Technology, Incorporated | Research and Development Intern

Las Piñas City, NCR, Philippines

Completed 300 hours of On-the-Job Training

- Integrated a Raspberry Pi SMS communication system to enable device-to-device data transfer with the use of GSM Modules.
- Diagnosed and resolved communication failures in IoT projects through systematic hardware-software debugging, reducing errors in systems and improving system reliability during testing.
- Produced technical documentation covering wiring configurations, testing procedures, and user manual setup for the prototype for ease of understanding within the development team

## TECHNICAL PROJECTS

### OmniTxt Lite SMS Communication System | 2026

Research and Development Specialist | IoT Technology, Inc.

- Contributed to testing and quality assurance of OmniTxT, a Raspberry Pi-based multi-port GSM gateway to enable SMS communication up to 8 simultaneous GSM modems. Designed to meet the Philippine NTC compliance requirements for high-throughput bulk messaging.
- Troubleshoot GSM module connectivity, signal instability, and serial communication issues using AT command diagnostics across multiple Raspberry Pi system nodes.
- Verified reliable operation across all system nodes, including parallel multi-modem dispatch, per-modem inbox polling, and a failover system, contributing to a production-ready gateway capable of serving up to 8 active SIM cards or more, across major Philippine carriers simultaneously.

### Active Directory Home Lab Environment | 2026 (Ongoing)

System Administrator Home Lab Project

- Deployed and configured a Windows Server 2022 Domain Controller in a virtual environment, establishing a working enterprise network for hands-on systems administration practice.
- Implemented Group Policy Objects to enforce security standards including account lockout, password complexity, and access restrictions, reducing simulated unauthorized access risks across the domain
- Configured shared folder permissions using Share and NTFS layers with group-based access control, then automated network drive mapping via GPO with Item-Level Targeting to ensure role-appropriate access.
- Simulated real IT helpdesk scenarios including password resets, account lockouts, and account recovery, documenting each process to reflect standard enterprise support workflows.

### CISCO Packet Tracer Network Lab | 2026 (Ongoing)

- Built and tested basic network topologies using Cisco Packet Tracer, connecting end devices through switches.
- Configured static IP addressing and verified device connectivity using ping and ICMP testing.
- Used Simulation Mode to visualize network traffic and understand how data moves across a network.

## EDUCATION

**STI College Las Piñas** | Bachelor of Science in Computer Engineering | 2022 – 2026

## CERTIFICATION

- AWS Academy Cloud Foundations (2025)
- CISCO Hardware Basics Training Certification (2026)
- CISCO Introduction to Cybersecurity Training Certification (2026)
- CISCO Ethical Hacker Certifications Training Certification (2026)
- ISC2 Candidate Certification (2026)
- Microsoft Azure-900 Fundamentals Certification (Ongoing 2026)